PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
rules = {"Common Cold":  {"sneezing", "runny_nose", "sore_throat"},
    "Flu":          {"fever", "cough", "body_pain", "fatigue"},
    "Malaria":      {"fever", "chills", "sweating", "headache"},
    "Dengue":       {"high_fever", "rash", "joint_pain", "headache"},
    "Typhoid":      {"fever", "stomach_pain", "weakness", "headache"},
    "COVID-19":     {"fever", "dry_cough", "loss_of_smell", "breathing_difficulty"},
}
def diagnose(symptoms):
    symptoms = set(symptoms)
    scores = []
    for disease, needed in rules.items():
        match = symptoms & needed
        if match:
            scores.append((len(match) / len(needed), disease, match))
    scores.sort(reverse=True)
    return scores
def show(symptoms):
    print("Symptoms:", sorted(symptoms))
    results = diagnose(symptoms)
    if not results:
        print("  No disease matched. Consult a doctor.\n")
        return
    for score, disease, match in results[:2]:
        print(f"  {disease}: {score*100:.0f}% match (matched: {sorted(match)})")
    best = results[0]
    if best[0] == 1.0:
        print(f"  -> Diagnosis: {best[1]}\n")
    else:
        print(f"  -> Possible: {best[1]} (consult a doctor)\n")
show({"fever", "cough", "body_pain", "fatigue"})
show({"sneezing", "runny_nose"})
show({"fever", "chills", "headache"})
```

OUTPUT:

```
>>>
================== RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ==================
Symptoms: ['body_pain', 'cough', 'fatigue', 'fever']
  Flu: 100% match (matched: ['body_pain', 'cough', 'fatigue', 'fever'])
  Typhoid: 25% match (matched: ['fever'])
  -> Diagnosis: Flu

Symptoms: ['runny_nose', 'sneezing']
  Common Cold: 67% match (matched: ['runny_nose', 'sneezing'])
  -> Possible: Common Cold (consult a doctor)

Symptoms: ['chills', 'fever', 'headache']
  Malaria: 75% match (matched: ['chills', 'fever', 'headache'])
  Typhoid: 50% match (matched: ['fever', 'headache'])
  -> Possible: Malaria (consult a doctor)

>>>
```

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